

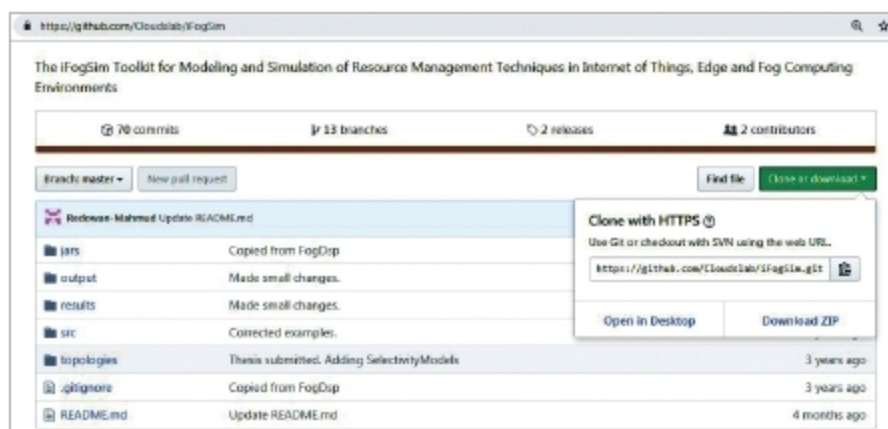


Figure 2: Downloading iFogSim

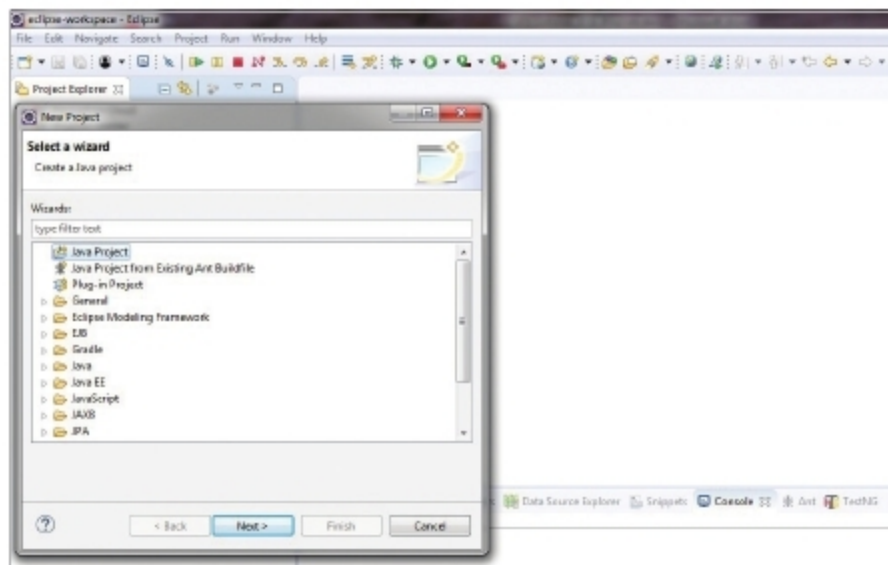


Figure 3: Creating a new project in the Eclipse IDE

## Installing iFogSim

The iFogSim library can be downloaded from the URL <https://github.com/Cloudslab/iFogSim>. This library is written in Java, and therefore the Java Development Kit (JDK) will be required to customise and work with the toolkit.

After downloading the compression toolkit in the Zip format, it is extracted and a folder *iFogSim-master* is created. The iFogSim library can be executed on any Java based integrated development environment (IDE) like Eclipse, Netbeans, JCreator, JDeveloper, jGRASP, BlueJ, IntelliJ IDEA or JBuilder.

In order to integrate iFogSim on an Eclipse IDE, we need to create a new project in the IDE (as shown in Figures 3 and 4).

## Simulating networks using iFogSim

Once the library is set up, the directory structure of iFogSim can be viewed in the Eclipse IDE in *Project Name* -> *src*.

There are numerous packages with Java code for different implementations of fog computing, IoT and edge computing.

To work with iFogSim in the graphical user interface (GUI) mode, there is a file called *FogGUI.java* in *org.fog.gui.example*. This file can be directly executed in the IDE, and there are different cloud and fog components that can be imported in the simulation working area as shown in Figure 5.

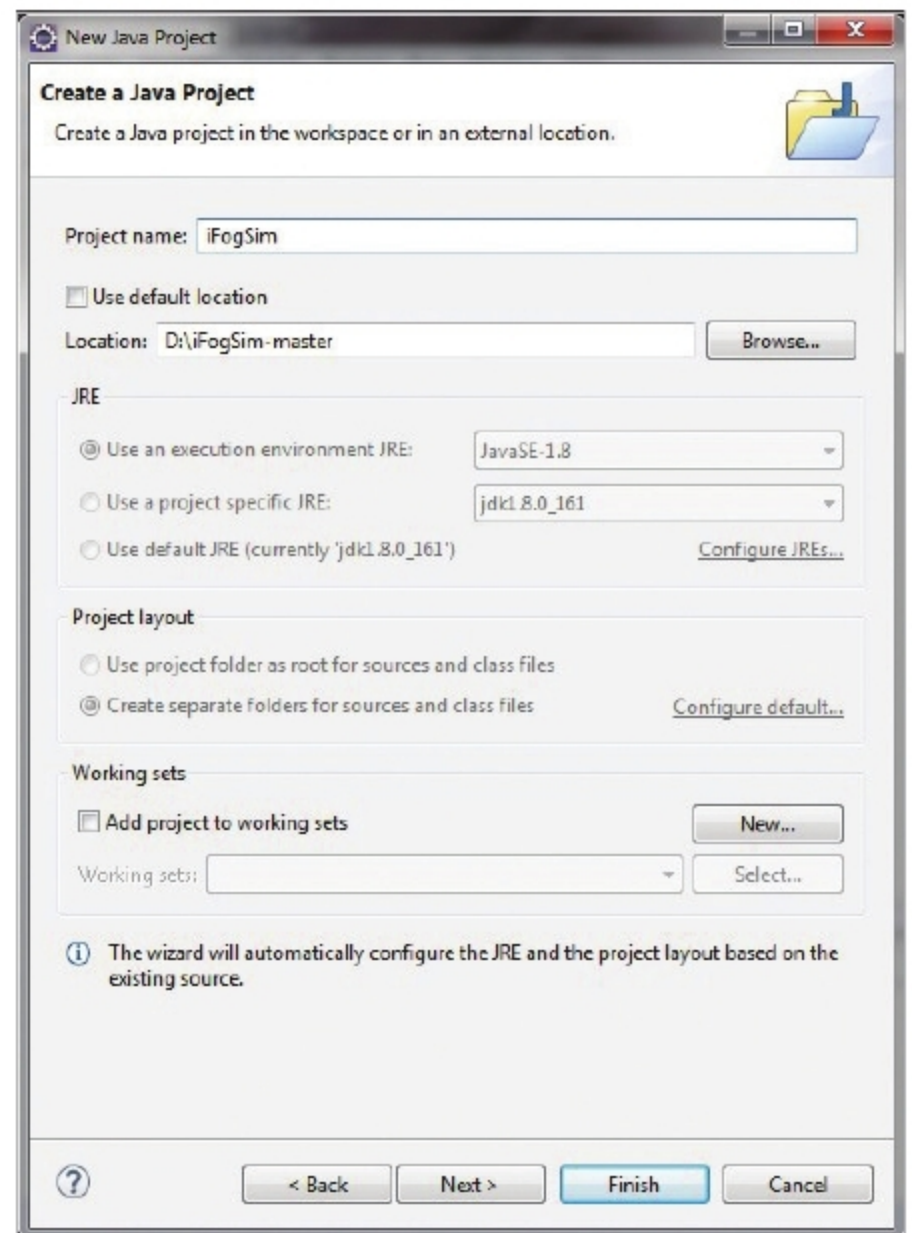


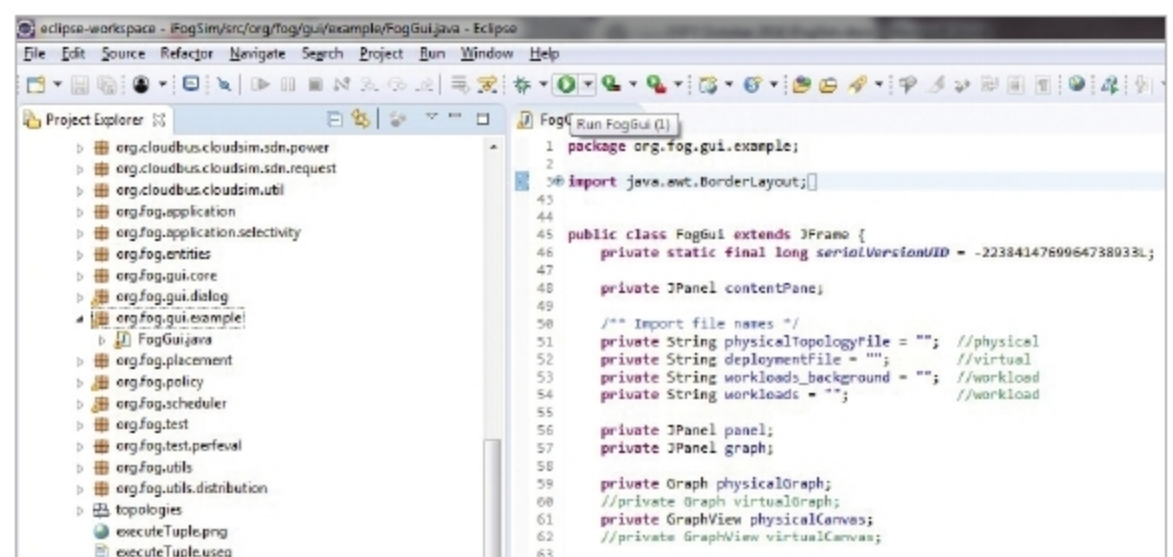
Figure 4: Integrating iFogSim with a project in the Eclipse IDE

In *Fog Topology Creator*, there is a *Graph* menu, where there is the option to import the topology (Figure 6).

On execution, the output can be viewed in the console of the Eclipse IDE.

In iFogSim, there are different scenarios for multiple applications that can be simulated; these include software defined networking (SDN) and its integration with cloud and fog computing.

In the base installation of iFogSim, there are a number of case studies that have been implemented and programmed. For example, there is a case study

Figure 5: Opening *FogGui.java* in Eclipse